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Sustainable Development and the Financial System: Society's Perceptions About Socially Responsible Investing

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ABSTRACT
The debate surrounding the financial needs of investors and the impact on society of investment is considered to be an important research topic due to the growth of socially responsible financial markets. The main objective of this research is to study society's perception about socially responsible investing (SRI) and to identify investor's preferences regarding environmental, social and governance criteria, their real-life investment needs and the most relevant sustainable financial products.

To examine society's perception of SRI, we conducted a field survey among Spanish investors. The results show that SRI is at an early stage and Spanish investors need more exact information regarding social, environmental, and governance criteria in order to invest in socially responsible companies and products.

This paper offers some guidelines that could be used by Spanish institutions, managers and investors and by foreign managers when approaching the Spanish market, in order to promote the growth of socially responsible financial products. Copyright © 2012 John Wiley & Sons, Ltd and ERP Environment.

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Introduction

SOCIALLY RESPONSIBLE INVESTING (SRI) IS A FINANCIAL INVESTMENT PROCESS THAT TAKES INTO ACCOUNT SOCIAL, ENVIRONmental and corporate governance impacts and/or investment in the community – as well as shareholder activism (SIF, 2009). Decisions about SRI have been studied in the economic and financial literature in recent years from various perspectives (Anand and Cowton, 1993; Williams, 2007) in an effort to establish the determinants of these investment choices. According to Nilsson (2009) it is reasonable to assume that motivations for investing in SRI differ depending on how the individual investor relates to the financial and social, environmental or ethical elements of the service.

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In recent years, SRI has become a key factor for institutional investors and is especially relevant for individuals. Investors have begun to consider and to incorporate ecological and socio-economic criteria in their investments. The European SRI market has increased from €2.7 trillion in 2007 to €5 trillion at the end of 2009; a growth of about 87% over 2 years (Eurosif, 2010). This growth can be partly attributed to changes in regulation regarding the disclosure of social, environmental and ethical information by pension funds and listed companies (Renneboog *et al.*, 2008) and the success of the United Nations Principles for Responsible Investment (Capelle and Monjon, 2012) which had more than 1087 signatories world-wide by July 2012.

However, the fundamental objective of both private and institutional investors is to maximise the expected rate of return on their investment portfolios (Koellner *et al.*, 2005). There is a relationship between demand for SRI and perception of financial performance, so lower future returns make fewer people want to invest in SRI (Lewis and Mackenzie, 2000a; Nilsson, 2008; Nilsson, 2009). On the other hand, by investing in SRI, investors receive returns that are not in the form of financial gain (Beal *et al.*,2005; Nilsson, 2009).

SRI is considered a very important topic in research due to the growth of financial markets. On the one hand, the inclusion of environmental criteria in the investment decisions of investors is making corporate behaviour more sustainable (Brammer and Pavelin, 2006). Firms and investors recognize that investing in accordance with the principles of sustainability can create long-term value (Bebbington, 2001) and improve firms' performance (Aslaksen and Synnestvedt, 2003). In this sense, better management of the impacts of corporate products, services, processes and other activities on various corporate stakeholders (including customers and investors) can improve both corporate sustainability performance and financial performance (Epstein and Widener, 2011). On the other hand, investor behaviour has become one of the most important drivers encouraging Western governments to introduce regulatory initiatives to stimulate SRI.

In Spain in the last few years a potential demand has appeared for products and services that could be defined under socially or environmentally responsible parameters. This notable rise could be caused in part by the appearance of funds and financial indices that use social responsibility selection criteria, such as the FTSE4Good IBEX index.

Under this assumption, one of the key aspects is to test whether these sustainable financial products take into consideration the main ethical, social and/or environmental perceptions and demands of consumers and investors. In addition, a number of factors have been identified as influencing the demand for SRI products (gender, education, occupation, etc.). To develop the Spanish SRI market these factors should be taken into account in the design of these sustainable financial products. In this context, the SRI market will not take off in Spain if there is not a constant demand from investors (retail and institutional). There is a need for an increased consciousness of their power in the market economy, and their ethical, social and environmental perceptions and demands need to be communicated to business. This paper aims to address this gap. In order to accomplish this, the objective of this article is to study society's perceptions about SRI – and to identify investors' preferences as regards environmental, social, governance criteria (ESG criteria), their real-life investment needs and the most relevant sustainable financial products, with special attention to the Spanish market. A questionnaire was developed to obtain the necessary information.

This paper is structured as follows: the Introduction is followed by a brief explanation of the Spanish SRI market. Subsequently, we show the methodology used and the results obtained. The final section provides the main conclusions.

The Socially Responsible Investment Market in Spain

SRI is interpreted in a different way depending on geographical area, with distinct characteristics linked to the particulars of each financial market and the legal and cultural framework of each country (Cañal *et al.*, 2009)

The main feature of the SRI market in Spain is the difference between the institutional market, the core market for SRI in Spain, and the retail market. According to a report published by the Socially Responsible Investing Observatory (ESADE, 2010) assets invested in SRI (pension funds and investment funds) in Spain at 31 December 2009 totalled €14,053,875,000 (10% more than 2008); and the total number of investors with SR criteria was

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814,766 (up 0.229% on the previous year). Some €13,919,368,000 of the total assets invested in SRI (representing 99% of total SRI market) was invested in the institutional market and primarily consisting of occupational pension funds. Assets invested in the retail market totalled €134,507,000, or 1% of the total. This fact emphasises the importance of institutional socially responsible investors in Spain, especially with respect to occupational pension funds, despite the absence of legislative measures to encourage SRI.

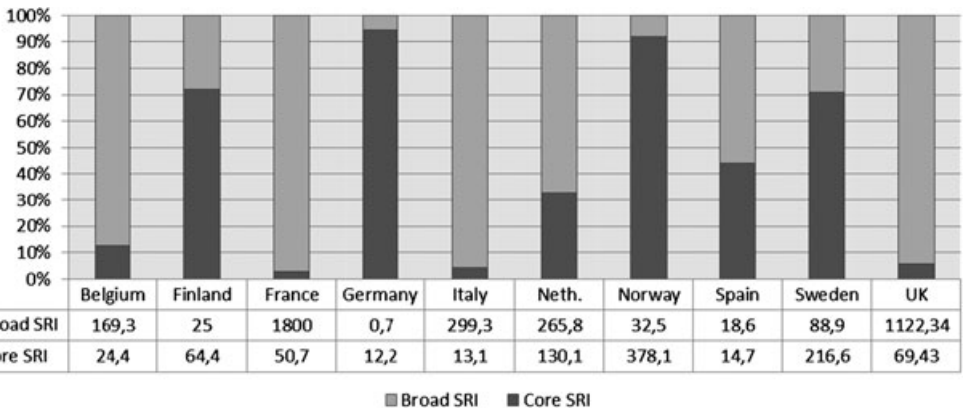
Moreover, compared with other countries, SRI is still in its early stages of development in Spain (Figure 1). According to the 'European SRI study' elaborated by Eurosif in 2010, for Core SRI, Norway, the Netherlands and Sweden show the fastest growth. Core SRI consists of ethical exclusions and positive screens (including best-in-class and pioneer screening). These strategies are often combined, especially in Switzerland, France and the UK. The broad SRI segment comprises simple exclusions (including norms-based screening), engagement and integration. France and the UK have had a remarkable increase in its broad SRI. In Spain, simple screening continues to be the most commonly used practice.

The Spanish financial market has several differences compared with other countries (Calvo *et al.*, 2005; Valor and De La Cuesta, 2007; Eurosif, 2010): (i) the recent history of mutual funds; (ii) the limited development of equity funds; (iii) the limited development of pension funds; (iv) engaging directly with companies is still not an integral part of Spanish sustainable investment practice; (v) lack of information about the available SRI offer in Spain and; (vi) the lack of trust about the ethical management of funds.

SRI was first discussed in Spain in the late 1990s when the first socially responsible investment funds (SR funds) were launched. Specifically, the first Spanish-based SR fund was launched in 1997 – the *Iber Fondo 2020 Internacional* – which included companies whose investments were aligned with the views of the Catholic Church. In the same year, the *Arco Iris de Ahorro Corporación* launched an investment fund for environmentally friendly technology companies. Two years later several more SR funds appeared on the Spanish market.

In November 1999, the Spanish Association of Investment and Pension Funds (INVERCO) issued guidance on the use by mutual funds of the terms 'ethical', 'ecological' and any other term regarding social responsibility. This guidance was approved by the Spanish Securities and Investments Board (CNMV) and its objective was to define the criteria that investment funds must meet to include these terms in their names. To standardise this situation and give greater transparency to ethical fund policies, the Spanish Standardisation and Certification Association (AENOR) published in 2002 a standard entitled: 'Requirements for Ethical and Socially Responsible Financial Instruments' (PNE 165001 EX) that specifies the requirements for ethical and socially responsible financial instruments created and sold by any organisation legally permitted to do so (Muñoz *et al.*, 2004). Regulation 31 of the Sustainable Economy Act 3/2011 (2011) (Regulations for the Dissemination of Information on Pension Fund Investment Policies) states that further regulations will be issued regarding the dissemination of information on ESG criteria in the investment policy of pension funds.

In addition to SR funds, other mechanisms are available for investors wishing to implement sustainability in their decision-making processes: sustainability rating agencies and sustainability indices. Sustainability indices facilitate the selection of investments – but restrict the investor's choice of socially responsible criteria.



	Belgium	Finland	France	Germany	Italy	Neth.	Norway	Spain	Sweden	UK
Broad SRI	169,3	25	1800	0,7	299,3	265,8	32,5	18,6	88,9	1122,34
Core SRI	24,4	64,4	50,7	12,2	13,1	130,1	378,1	14,7	216,6	69,43

Figure 1. The European socially responsible investment (SRI) market 2009 (billion €) (based on the report elaborated by Eurosif, 2010).

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The first Spanish sustainability stock index – the FTSE4Good IBEX index – was launched in 2008 and includes medium and large companies from the Spanish stock exchange. The index is part of the FTSE4Good family of indices and follows established inclusion and exclusion criteria.

Methodology and Sample

The Survey

To examine the public perception about SRI criteria and the real-life needs of investors we conducted a self-administered survey.

Self-administered survey is a quantitative research method that does not requires the use of an interviewer in administering the survey. Respondents read the questionnaire and record their responses themselves.

Specifically, we have designed a self-administered survey because: (i) this approach enables the study of a large number of both objective and subjective aspects that are not directly observable; (ii) information is obtained from a wide range of people in different geographical areas and within a relatively short period of time; (iii) the approach facilitates comparison between responses; (iv) the questionnaire reduces the bias that might result from personal characteristics of interviewers, etc. The disadvantages include the relatively low response rate and absence of personal contact with the respondents (Bourque and Fielder, 2002).

The questionnaire that was used was designed and tested in 2010. Five hundred and twenty-five individuals were surveyed from around Spain between April 2010 and November 2010. However, the final sample was reduced to 66% of the valid responses. This reduction was made because many respondents did not answer some specific questions. A lack of awareness of the product by many respondents was observed – this phenomenon could be the result of the poor development of SRI in Spain.

Research using small sample sizes is not uncommon in this field of study, for example: Bigné *et al.* (2005) studied the perception of corporate social responsibility (CSR) in four nations with sample sizes of fewer than of 155 subjects per nation (418 total responses); Anand and Cowton (1993) explored socially responsible investor behaviour with a sample of 125 observations from the EIRIS database; Lewis and Webley (1994) observed that the majority of investors showed an attitude revealing a positive predisposition to make social and environmental investments using a questionnaire given to 100 investors; McLahan and Gardner (2004) provided a comparative examination of 55 conventional and 54 socially responsible investors, with the aim of identifying the factors that lead investors to choose SRI products.

The questionnaire for the study was made up of 28 items and included five sections: (i) general questions about the person completing the survey; (ii) questions related to the destination of the respondent's savings; (iii) questions related to SRI; (iv) questions related to socially responsible funds; and (v) questions related to sustainability indices and sustainability rating agencies. Various references were used in designing the questionnaire – such as those issued by Balaguer *et al.* (2008) to managers, Benijts (2010), the previously cited Bigné *et al.* (2005), Economistas sin Fronteras (ESF) and UNED (2007), or Valor *et al.* (2008) – that questioned Spanish individuals and financial consultants about their investment strategies, their investment preferences and the perceived obstacles for the development of SRI.

The Sample

The final sample was composed of 345 respondents; of these we considered only 20 respondents as socially responsible consumers. According to Roberts (1995), we defined the socially responsible consumer as 'one who purchases products and services perceived to have a positive (or less negative) influence on the environment or who patronizes businesses that attempt to effect related positive social change'.

The results for the profile of socially responsible investors are not unanimous. However, many studies show that socially responsible investors tend to be: university-educated women (Tippet and Leung, 2001; Schueth, 2003; Williams 2007); university-educated men (Haigh, 2008); in late middle-age (Lewis and McKenzie, 2000a,2000b, ESF and UNED, 2007); and with an above-average income (Lewis and McKenzie, 2000a,2000b, Haigh, 2008). This

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Variables		Total sample ($n = 345$)	Socially responsible consumers[1]
Gender	Male	53.3%	4.9% of all men
	Woman	46.7%	6.8% of all women
Age (years)	18–24	9.9%	5.8% of total
	25–34	39.7%	6.6% of total
	35–45	27.2%	5.3% of total
	45–54	8.7%	3.4% of total
	55–64	9.9%	8.8% of total
	65 or older	4.6%	0
Education	No education	0.3%	0
	Primary school	3.8%	7.6% with no education
	Secondary school	18.8%	4.6% secondary school
	Degree	16.8%	8.6% with degree
	Post-graduate	60.3%	53.4% with graduate studies
Occupation	Student	16.5%	8.8% of total students
	Working	69%	4.6% of total workforce
	Homemaker	2.7%	0
	Retired	9.6%	12.1% of total pensioners
	Unemployed	2.3%	0
Monthly family household income	€0–1000	7.8%	7.4% of total
	€1001–1500	10.7%	16.2% of total
	€1501–2000	15.4%	9.4% of total
	€2001–2500	18.3%	1.6% of total
	> €2500	47.8%	3.6% of total
Percentage of income saved	0–10%	34.5	6.7% of total
	10–30%	37.7%	6.9% of total
	30–50%	18.5%	3.1% of total
	50–70%	7%	4.2% of total
	> 70%	2.3%	0%

Table 1. Profile of sample rates
[1]Any individual buying a socially responsible financial product is considered to be a socially responsible consumer.

lack of consensus may be due to the size of the samples, the use of differing methodologies, differing survey designs, and the varying nature of the nations studied, etc.

Coinciding with the literature, in our study socially responsible investors are mostly women, middle-aged, university-educated, and with average incomes and retired (Table 1).

Results

The main results of the study are presented below. Firstly, the data are statistically analysed by applying percentages; secondly, the possible relationships between the variables are explored to better understand investors' decisions.

Statistical Analysis

Socially Responsible Investing

The decision-making process of a socially responsible consumer is more complex than the process for consumers of traditional financial products. This is because social and environmental criteria are sometimes more important than financial criteria (Cañal *et al.*, 2009). Table 2

Comparing our results to those of Valor *et al.* (2008) who published a study entitled 'Understanding Demand for Retail Socially Responsible Investments: a Survey of Individual Investors and Financial Consultants' – in which they

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1. Have you heard of SRI?						
		Yes		No	No answer	
$n=345$		54.2%		44.3%	1.5%	
$n=20$		100%		0%	0%	
2. Have you bought a socially responsible financial product?						
		Yes		No	No answer	
$n=345$		5.8%		87.2%	7%	
$n=20$		100%		0%	0%	
3. Which socially responsible financial product have you bought?						
	SRI funds	Bonds and other SR loans	Guarantee funds	Funds for micro-credits	SR current account and savings deposits	Other
$n=345$	1.4%	0.9%	0.6%	0.3%	2.3%	1.6%
$n=20$	25%	15%	10%	5%	40%	20%
4. Why have you chosen, or would choose, a financial product that includes social and environmental variables?						
	High returns	Security of investing in sustainable products	Advice from bank	Ethical or religious reasons	Social pressure for more responsible behaviour	Other
$n=345$	28.1%	39.7%	7%	23.3%	5.5%	14.8%
$n=20$	10%	40%	25%	20%	5%	15%
5. Why have you not chosen, or would not choose, a financial product that incorporates social and environmental variables?						
	Low returns	Doubts about relationship between ethics and returns	Failure to publicise the existence of SR products	Lack of support or endorsement by the state	Charity and donation	Other
$n=345$	10.7%	18.3%	50.4%	6.6%	5.5%	17.1%
$n=20$	10%	20%	45%	20%	0%	5%
6. Do you know what ethical banking means?						
		Yes		No	No answer	
$n=345$		31.3%		65.2%	3.5%	
$n=20$		70%		30%	0%	

Table 2. Responses to questions on socially responsible investing (SRI)

analysed the demand for SRI in Spain – we observe that SRI market will not take off in Spain. Although 54.2% of the sample recognise the term SRI (up 17.4% on the previous study) only 5.8% of the total sample has ever purchased a socially responsible financial product (up 3.04% on the previous study).

The most frequently purchased products are SR current account and savings deposits (2.3% of the sample and 40% of SR investors). The lack of information about the existence of these products seems to be the most common cause for non-purchase (54.4% of the sample and 45% of SR investors). The security implied in investing in sustainable products is the main reason for purchasing a SR financial product (39.7% of the sample and 40% of SR investors). In addition, only 31.3% of the respondents had heard of ethical banking. Ethical banks are concerned with the social and environmental impacts of their investments and loans.

SRI Funds

The rapid growth of socially responsible mutual funds is certainly linked to the increased level of interest in socially responsible investing observed in recent years, and SRI mutual funds are probably the most prominent financial instrument in the field of ethical finance (Bassso and Funari, 2010). However, although there are many methods for assessing financial performance, the assessment of non-financial performance is rather underdeveloped. Consequently, fund managers are not able to set up standards for non-financial performance, and thus they are unable to account for this aspect to investors and their stakeholders (Koellner *et al.*, 2005).

One of the market problems is that Spanish fund managers have not developed appropriate strategies for publicising SR funds – and specifically in the financial sector (IPES, 2002). According to the research of Barreda

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et al.(2011), when information on the degree of SR of the funds was disclosed, overall investment in the SR fund was higher. In our study, less than 12.8% of the sample had received information about investing in SR funds.

Reviewers positively evaluated SR products for profitability (55.1%), followed by social aspects (22.6% of the sample), ethical aspects (15.9%), and environmental aspects (6.4% of the sample) (Table 3).

SRI mutual funds apply a combination of various types of screens: negative screens and positive screens. The negative screens consist of negative (exclusion) criteria that exclude certain investments; while positive screens identify those companies with best behaviours. The use of positive screens is often combined with a 'best in class' approach.

There is no consensus in the literature regarding the social aspects that most concern investors (Tippet and Leung, 2001). These aspects vary depending on the legal framework (Cañal *et al.*, 2009) as well as the cultural and social issues in each country (Katz *et al.*, 2001; Sirmon and Lane, 2004) and social norms (Hong and Kacperczyk, 2009).

In Nordic and Anglo-Saxon nations, including the Netherlands, environmental and ecological values are highlighted. Anglo-Saxon countries, especially the UK, support community development and employment

7. Have you received information about SRI funds?					
	Yes		No		No answer
$n=345$	12.8%		85.2%		2%
$n=20$	70%		30%		0%
8.If you were to invest in a SRI fund what would you value most?					
		Returns	Ethical aspects	Social aspects	Environmental aspects
$n=345$		55.1%	15.9%	22.6%	6.4%
$n=20$		35%	20%	25%	20%
9. What is the most valued or positive criterion for you when deciding to invest in a SRI fund?					
	$n=345$	$n=20$		$n=345$	$n=20$
Equal opportunities	7%	10%	Pollution control	4.3%	5%
Recycling	0.6%	0%	Workplace safety	4.3%	15%
Conserve energy and natural resources	16.5%	20%	Training help	3.5%	10%
Information transparency	21.7%	5%	Corporate investment	0.9%	5%
Social solidarity	31.6%	25%	Others	9.6%	5%
10. What is the most excluding or negative criterion when deciding to invest in a SRI fund?					
	$n=345$	$n=20$		$n=345$	$n=20$
Gambling-related activities	0.6%	0%	Nuclear production	2.3%	5%
Activities relating to tobacco and alcohol	1.2%	5%	Activities that abuse the environment	5.5%	5%
Practices related to embryo research	0.6%	0%	Activities related to pornography	2.9%	5%
Abortion practices	7.8%	0%	Animal testing	1.4%	5%
Activities that abuse and human and labour rights	36.5%	35%	Support for repressive or dictatorial regimes	13.9%	20%
Lack of transparency in business practices	12.2%	0%	Genetic manipulation	1.2%	0%
Activities related to armaments	7.5%	10%	Others	6.4%	10%
11.In your opinion, when compared with traditional investment funds, a SRI fund is:					
	Riskier	A little riskier	Similarly risky	A little less risky	Much less risky
$n=345$	6.4%	11.6%	62%	11.3%	8.7%
$n=20$	10%	10%	50%	15%	15%
12. In your opinion, when comparing SR in the long run with traditional investment funds, a SRI fund offers:					
	Much better returns	Better returns	Similar returns	Lower returns	Much lower returns
$n=345$	1.2%	7.8%	50.4%	35.4%	5.2%
$n=20$	0%	10%	50%	35%	5%

Table 3. Results of questions about socially responsible investment (SRI) funds

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initiatives. In the Mediterranean area, the characteristic values are social solidarity, social action and the social inclusion of marginalised groups – as well as religious values. Countries between the Mediterranean and Nordic regions (France, Belgium and Austria) combine the values of both areas and give special importance to worker rights and trade union issues (Cañal *et al.*, 2009).

The exclusion criterion most valued by respondents is activities that abuse human and labour rights (36.5% of the sample and 35% of SR investors), while the least valued exclusion criteria are gambling-related activities and practices related to embryo research (0.6%) (Table 3).

The most highly valued positive criterion for most respondents was a commitment to social solidarity (31.6% of the sample and 25% of SR investors); and the least valued was recycling (0.6% of the sample).

Comparing these results with those of Valor *et al.* (2008), in both studies the exclusion criterion most valued by respondents is activities that abuse human and labour rights and the most highly valued positive criterion for most respondents is related to social aspects.

The criteria most valued by Spanish investors scarcely match the declared values and exclusion criteria of the main SR funds registered with the CNMV in 2009. Only five of the 11 funds analysed in Table 4 explicitly exclude activities that abuse human and worker rights. Just two funds have a positive criterion regarding commitment to social solidarity.

Fund	Negative criteria	Positive criteria
Bankinter Gestión Ambiental FI	None mentioned	▪ Activities to limit climate change ▪ Correct agricultural use of natural resources ▪ Clean energy ▪ Use of biotechnology
BNP Paribas Fondo Solidaridad, FI	Companies abusing human development, justice and world peace. These criteria are: ▪ violations of human rights ▪ practices dangerous to public health ▪ arms or military work – provided these activities exceed 10% of turnover ▪ highly polluting industries that harm the environment ▪ activities related to pornography and sexual exploitation	Positive assessment criteria include: ▪ job security ▪ social support ▪ worker education and training ▪ employment of the young, women and the socially excluded ▪ environmental policies ▪ transparency ▪ social solidarity, reconstruction and development
Compromiso Fondo Ético	Excludes companies that: ▪ make arms ▪ discriminate between employees and abuse human rights ▪ produce or distribute tobacco, alcohol, or other harmful products	Positively values companies that: ▪ are environmentally certified ▪ encourage job security and hire the disabled
Espirito Santo Ético y Solidario, FI	Exclude investments in companies whose income and profits come from: ▪ arms making ▪ alcohol ▪ tobacco	Investment in companies that: ▪ encourage local development or community development in developing nations ▪ recognise worker rights ▪ voluntarily contribute to social welfare

(Continues)

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Fund	Negative criteria	Positive criteria
	▪ harmful products ▪ abuses of worker rights ▪ harming the environment	▪ contribute to transparency in all activities ▪ incorporate good governance procedures ▪ encourage worker rights and job security ▪ encourage training and development ▪ encourage technologies that are environmentally sustainable ▪ create jobs for the disabled
Foncaixa Cooperación Socialmente Responsable Europa FI	Excludes: ▪ makers of tobacco and arms ▪ owners and operators of nuclear power plants ▪ companies mining and using uranium	Investments aimed at companies that meet CSR recognised standards, including the FTSE4GOOD Index Europe (corporate governance, social criteria, environmental protection, investor relations and human rights)
Foncaixa Privada Fondo Activo Ético, FI, de Caja de Ahorros y Pensiones de Barcelona	Companies that violate fundamental human rights; the Pharmaceutical Code and/or the International Code of Marketing of Breast-milk Substitutes; make or distribute weapons; operate nuclear power plants; or manufacture, sell or distribute tobacco	An ethics commission values as strengths and weaknesses relative to corporate governance, the environment, human rights, and stakeholders (topics of interest for shareholders)
Fondo Solidario Pro-UNICEF, FI (Caja Madrid)	The fund does not invest in: ▪ arms makers ▪ alcohol producers ▪ tobacco producers Companies are excluded if 25% of sales are from one of the above sectors	None mentioned
Fonpenedes Etic i Solidari	Excluded sectors include: ▪ tobacco companies ▪ companies supplying nuclear power industry ▪ arms makers ▪ owners and operators of nuclear power plants	Inclusive criteria include: ▪ environmental management and climate change ▪ human and labour rights ▪ minimum work conditions from suppliers ▪ rejection of corruption
Santander Dividendo Solidario Europa, FI	Companies whose aims and ideals are unethical, such as making and selling weapons. Non-compliance with the Kyoto Protocol. Make harmful products, guilty of environmental pollution, irresponsible or offensive advertising	Inclusive criteria include: ▪ employment for young people and women ▪ environmental policies ▪ education and training of employees ▪ health and social security practices
Santander Responsabilidad Conservador, FI	Companies whose aims and operations work against the principles and values of the Catholic church, such as: ▪ those who investigate/manipulate human embryos and produce abortions ▪ have been reported for child labour practices, or have been convicted for discriminating against workers	None mentioned

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Fund	Negative criteria	Positive criteria
	▪ encourage violence through the production of toys and video games, or participate significantly in the capital of others that do so ▪ produce or sell substances that impoverish the ozone layer and favour climate change, or participate significantly in the capital of others that do so ▪ companies that violate WHO recommendations, or manufacture tobacco products or distilled spirits ▪ companies that have been found guilty of abusing market competition, consumer exploitation, violation of urban planning, corporate crime, etc	
Urquijo Inversión Ética y Solidaria	Companies that: ▪ harm integral human development, justice, and social peace ▪ abuse basic human rights ▪ harm public health ▪ make or sell arms ▪ make or sell tobacco ▪ experiment with human genes ▪ make products or sell services that harm human life and dignity ▪ exploit child labour ▪ harm the environment	Inclusion in the EIRIS database (Ethical Investment Research Services).

Table 4. Exclusion criteria and values of the principal Spanish-based and managed SR funds registered with the Spanish Securities and Investments Board (CNMV) in 2009 (source: own work).

Another major problem in the Spanish SRI market is that many investors are unaware that the returns on SR are the same as with any other fund in the same category, given that the management approach is the same. Some 40.6% of the respondents believe that financial returns on SR are less than traditional investment funds. In terms of risk, 18% of respondents believe that these funds are riskier than traditional funds. Spaniards' risk aversion limits SRI integration, since it decreases the percentage invested in equity (Valor *et al.*, 2008).

Most empirical results discussed in the literature suggest that SRI mutual funds produce better or similar returns than other funds (Statman, 2000; Geczy *et al.*, 2006; Bauer *et al.*, 2006; 2007; Fernández and Matallín, 2008). However, some empirical results support the opposite conclusion (Jones *et al.*,2008; Chang and Witte, 2010).

Sustainability Rating Agencies and Sustainability Indices

On the one hand, sustainability indices, in the same way as traditional stock market indices, are indicators of the price trends of the most representative shares in a stock market. However, in this case, the market is limited to socially responsible companies. These indices are also used as benchmarks in the management of socially responsible portfolios. While these indices help investors choose investments, they also restrict the application of an investor's own criteria regarding social responsibility.

According to the 'The value of transparency' elaborated by Foretica (2012), which compares the evolution of a sustainability index (MSCI World SRI series) to a traditional stock market index (MSCI AC World Index) from 2007 to 2012, the SRI index entailed a lower risk measured both in terms of Beta and Volatility. During the financial crisis (2008–2009), the SRI index forecasted lower drops along with lower volatility.

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On the other hand, sustainability rating agencies are the link between stakeholders and companies (Schäfer, 2005). It is clear that these agencies are growing due to two key factors: the expansion of the securities markets and the growth of regulation (Ferri and Liu, 2005) regarding the disclosure of social, environmental and corporate governance information.

Sustainability rating agencies study businesses and use their own research methodologies to make evaluations on ESG aspects. Normally, evaluations are made by using complex questionnaires and an analysis of public information sources. The criteria used by sustainability rating agencies are later reflected by the most prestigious sustainability indices. Table 5

Sustainability indices and ESG agencies are important sources of information for the socially responsible investor. However, these agencies remain little known by Spanish investors.

Only 26.1% of the sample has heard of sustainability indices. The best known index is the Dow Jones Sustainability Index (recognised by 16.2% of the sample and 40% of SR investors), followed by the FTSE4Good IBEX (7.8%), FTSE4-Good (7.5%), FTSE4 KLD 400 Social Index (1.7%), Ethibel Sustainability Index (1.4%), others (1.4%), and ASPI (0.6%). None had heard of the Calvert Social Index.

The percentage of respondents who understand something about the role of sustainability rating agencies is 17.7%. The best known sustainability rating agencies are: Accountability Rating (3.5% of the sample); other agencies (3.2%); EIRIS (2.9%); SAM (1.4%); ASSET4 (1.4%); Vigeo (1.2%); Research Oekom AG (0.9%); and KLD Research & Analytics (0.9%). The lack of familiarity with these agencies may be because agencies reveal little when explaining the evaluation criteria used, particularly those related to risk management. As a result, a direct analysis of information by investors is difficult (Escrig *et al.*, 2010).

Some 58% of respondents believe SRI will grow in popularity in Spain.

The Logistic Regression Model

To analyse the influence of gender, educational level and occupation in some answers we used a linear probability model, namely, the logistic regression model (Tabachnick and Fidell, 2007). This is a generalisation of the classic

13. Have you heard of stock indices for socially responsible investment?					
	Yes		No		No answer
$n=345$	26.1%		73%		0.9%
$n=20$	55%		45%		0%
14. Do you recognise any of the SRI stock indices in the financial market?					
	$n=345$	$n=20$		$n=345$	$n=20$
Ethibel Sustainability Index	1.4%	10%	ASPI (ARESE Sustainable Performance Indices)	0.6%	0%
Dow Jones Sustainability Index	16.2%	40%	FTSE KLD 400 Social Index	1.7%	5%
FTSE4Good	7.5%	20%	Calvert Social Index	0%	0%
FTSE4Good IBEX	7.8%	20%	Other	1.4%	0%
15. Have you heard of sustainability analysis agencies?					
	Yes		No		No answer
$n=345$	17.7%		79.1%		3.2%
$n=20$	30%		65%		5%
16. Do you recognise any of the sustainability analysis agencies in the financial market?					
	$n=345$	$n=20$		$n=345$	$n=20$
EIRIS	2.9%	5%	SAM	1.4%	5%
Oekom Research AG	0.9%	5%	ASSET4	1.4%	5%
Vigeo	1.2%	5%	KLD Research & Analytics Inc.	0.9%	0%
Accountability Rating	3.5%	10%	Other	3.2%	5%
17. What do you think will be the tendency in Spain regarding socially responsible investment?					
	Decrease		No change	Increase	No answer
$n=345$	2.6%		18%	58%	21.4%
$n=20$	0%		15%	75%	10%

Table 5. Results of questions on sustainability rating agencies and sustainability indices

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Dependent variable	Independent variable	Hypothesis	Test	Nature of the relation
Awareness_SRI	General level, education, occupation	H_1: Awareness_SRI depends on the general variables, education and occupation	Multivariate binary logistic regression	H_1 depends on gender, education and occupation
Purchase_SRI	General level, education, occupation	H_2: Purchase_SRI depends on the general variables, education and occupation	Multivariate binary logistic regression	None
Awareness_ethical banking	General level, education, occupation	H_3: Awareness_ethical banking depends on the general variables, education and occupation	Multivariate binary logistic regression	H_3 depends on education and occupation
Information_SRIfunds	General level, education, occupation	H_4: Information_SRIfunds depends on the general variables, education and occupation	Multivariate binary logistic regression	None
Evaluation_SRIfunds	General level, education, occupation	H_5: Evaluation_ SRIfunds depends on the general variables, education and occupation	Multinomial logistic regression	H_5 depends on gender and occupation
Criteria_positives	General level, education, occupation	H_6: Criteria_positives depends on the general variables, education and occupation	Multinomial logistic regression	H_6 depends basically on occupation
Criteria_negatives	General level, education, occupation	H_7: Criteria_negatives variable depends on the general variables, education and occupation	Multinomial logistic regression	H_7 depends basically on education and occupation
SRIfunds_risk	General level, education, occupation	H_8: SRIfunds_risk variable depends on the general variables, education and occupation	Ordinal logistic regression	None
SRIfunds_returns	General level, education, occupation	H_9: SRIfunds_return variable depends on the general variables, education and occupation	Ordinal logistic regression	None
Awareness_indices	General level, education, occupation	H_{10}: Awareness_índices variable depends on the general variables, education and occupation	Multivariate binary logistic regression	H_{10} depends on education
Awareness_agencies	General level, education, occupation	H_{11}: Awareness_agencies variable depends on the general variables, education and occupation	Multivariate binary logistic regression	H_{11} depends on education
Tendency_SRI	General level, education, occupation	H_{12}: Tendency_SRI variable depends on the general variables, education and occupation	Ordinal logistic regression	H_{12} depends on education and gender

Table 6. Hypotheses for investors

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linear regression model for categorical dependent variables. It is normally employed to predict whether an observation belongs to a particular group or to analyse the influence of a series of independent variables on the dependent variable (Cardone *et al.*, 2010). In our case, we have used a logistic regression model with the second purpose.

The main features include: (i) assumptions of normality and equality by variance–covariance matrices do not have to be met; (ii) the model is more robust than other techniques when these assumptions are not met; (iii) the use of continuous and categorical independent variables is enabled; (iv) direct statistical tests are included; (v) nonlinear effects can be incorporated and the model is useful for diagnosis (Hair *et al.*, 1999); and (vi) the model is widely used in observational, survey and experimental studies. Various studies have opted to use this model to analyse socially responsible relations and attitudes (Balaguer *et al.*, 2008; Tranter and White, 2001; Chan and Wickramasinghe, 2006).

In our study, we have used three types of logistic regressions based on the type of dependent variable analysed. These regressions are: multivariate binary logistic regression when the dependent variable has two categories; multinomial logistic regression when the dependent variable has more than two categories; and ordinal logistic regression when the order of dependent variable categories is fixed.

Hypotheses

Below we identify the attitude and factors that motivate Spanish investors in the investment process, and study the influence of gender [male (1) versus female (0)], education level [university (1) versus no university (0)], and occupation [employed (1) versus unemployed and non-actives (0)]. Because gender, education and occupation are such critical stratification variables, it is important to understand how they affect investment decision process.

Table 6 shows the hypotheses. The outcomes of the first analysis of the logistic regression test can be seen in the results column.

Results of the Model

Logistic regression calculates the probability of success over the probability of failure:

$$\text{probability (event)} = \frac{1}{1 + e^{-z}}.$$

The results of our analysis are in the form of an odds ratio and Wald test. The odds ratio is a measure of effect size (Westergren *et al.*, 2001), describing the strength of association or non-independence between two binary data values:

If the probability (success) < probability (failure), then odds (success) < 1.
If the probability (success) > probability (failure), then odds (success) > 1.

To test the significance of individual logistic regression coefficients for each independent variable (that is, to test the null hypothesis), we use the Wald test and odds ratio.

In the tables of results the Wald statistics for testing the significance of the explanatory variables are given in the column 'Wald'. The Wald statistics are approximately χ^2 (chi-squared) distributed with the degrees of freedom given in the column 'df'. The *P*-value for the test statistics are given in the column 'Sig'.

The predicted odds ratio is output in SPSS in the column 'Exp(B)'. We have converted odds ratios to probabilities to know the odds ratio predicted by the model. Probability ranges from 0 and 1. Odds ratios range from 0 and positive infinity:

$$\text{probability} = \frac{\text{odds ratio}}{1 + \text{odds ratio}}.$$

After checking the hypotheses H_1, H_2, H_3, H_4, H_{10} and H_{11}which are related to the level of awareness and information, it appears that the gender variable does not influence SR investment decisions except in $H_{1.}$ In this case, the SRI awareness of men, because 'male' is our reference category, is 0.621 times higher than SRI awareness of women. In other words, our model predicts that 38.3% of men would recognise the term SRI (Table 7).

The level of education variable is statistically significant ($P < 0.05$) in H_1, H_3, H_{10} and H_{11}. This means that it is 0.299/1 more likely that a people with a high level of education (university studies being our reference category) have a better knowledge of SRI and it is 0.313/1 more likely that more highly educated people have a better knowledge of ethical banking (Table 7).

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			B	SE	Wald	df	Sig.	Exp(*B*)	Probability
H_1	Step 1[1]	Gender(1)	−0.477	0.224	4.534	1	0.033	0.621	0.383
		Education_level(1)	−0.853	0.279	9.368	1	0.002	0.426	0.299
		Occupation(1)	0.610	0.256	5.659	1	0.017	1.841	0.648
		Constant	0.404	0.168	5.788	1	0.016	1.497	0.600
H_2	Step 1[1]	Gender(1)	0.332	0.469	0.500	1	0.479	1.393	0.582
		Education_level(1)	−0.445	0.599	0.551	1	0.458	0.641	0.391
		Occupation (1)	0.705	0.483	2.128	1	0.145	2.024	0.669
		Constant	−3.124	0.391	63.884	1	0.000	0.044	0.042
H_3	Step 1[1]	Gender(1)	−0.199	0.239	0.690	1	0.406	0.820	0.451
		Education_level(1)	−0.788	0.330	5.683	1	0.017	0.455	0.313
		Occupation(1)	−0.457	0.278	2.705	1	0.100	0.633	0.388
		Constant	−0.414	0.172	5.772	1	0.016	0.661	0.398
H_4	Step 1[1]	Gender(1)	0.096	0.327	0.087	1	0.768	1.101	0.524
		Education_level(1)	−0.778	0.474	2.690	1	0.101	0.460	0.315
		Occupation (1)	0.194	0.359	0.291	1	0.590	1.214	0.548
		Constant	−1.885	0.246	58.780	1	0.000	0.152	0.132
H_{10}	Step 1[1]	Gender(1)	−0.186	0.251	0.550	1	0.458	0.830	0.454
		Education_level(1)	−0.856	0.352	5.903	1	0.015	0.425	0.298
		Occupation(1)	0.136	0.279	0.237	1	0.627	1.145	0.534
		Constant	−0.834	0.182	20.957	1	0.000	0.434	0.303
H_{11}	Step 1[1]	Gender(1)	0.026	0.288	0.008	1	0.927	1.027	0.507
		Education_level(1)	−1.030	0.437	5.556	1	0.018	0.357	0.263
		Occupation(1)	0.211	0.317	0.444	1	0.505	1.235	0.553
		Constant	−1.433	0.213	45.438	1	0.000	0.239	0.193

Table 7. Hypotheses tested through binary logistic regression
B is the coefficient for the constant and Exp(B) is the exponentiation of the coefficients.
[1]Extraction to SPSS.

The occupation variable is significant in H_1 and H_3 with some degree of relaxation ($P < 0.10$)[1] (Table 7). This means, that employed investors (our reference category) are 0.648/1 times more likely to recognise the term SRI than unemployed investors(Table 7).

The hypotheses H_2 and H_4 formulated in the previous section cannot be accepted for these data.

We can therefore say that the degree of awareness of SRI depends mainly on the education variable – as higher education generates a more informed society.

Hypotheses H_5, H_6 and H_7 are related to the evaluation of SRI funds and the positive and negative criteria for making an investment in this type of fund. For these hypotheses, we obtained the following results.

> H_5: On the one hand, the variable 'gender' has a positive and statistically significant impact – with some degree of relaxation ($P < 0.10$) – on the probability of deciding to invest in SRI funds according to social aspects. That is, women are 0.580 times more likely to invest according to social aspects than men. On the other hand, gender and occupation are crucial when considering environmental aspects. In fact, it is 26.8% more likely that women will invest according to environmental aspects than men (Table 8).

[1]Statistical significance ($P < 0.05$) would be an overly restrictive term, but a certain degree of relationship does exist (for example, $P < 0.10$). The looseness of these recommendations is because a standard that is as restrictive as $P < 0.05$ may lead to a failure to include in the model covariates with a weak association with the dependent variable in isolation. However, these covariates may prove strong predictors for the dependent variable when read in conjunction with other covariates. Research using statistical significance ($P < 0.10$) is not uncommon in the literature such as Kale *et al.* (2002) and Co and Chew (1997).

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		B	SE	Wald	df	Sig.	Exp(*B*)	Probability
Social aspects	Intersection	−0.813	0.201	16.366	1	0.000		
	[Gender = 0]	−0.545	0.279	3.816	1	0.051	0.580	0.367
	[Education_level = 0]	0.354	0.323	1.201	1	0.273	1.424	0.587
	[Occupation = 0]	0.292	0.305	0.914	1	0.339	1.338	0.572
Environmental aspects	Intersection	−2.236	0.352	40.372	1	0.000		
	[Gender = 0]	−1.004	0.495	4.107	1	0.043	0.367	0.268
	[Education_level = 0]	0.392	0.520	0.570	1	0.450	1.480	0.597
	[Occupation = 0]	1.038	0.479	4.691	1	0.030	2.823	0.738

Table 8. Test for H_5: Evaluation_SRI
B is the coefficient for the constant and Exp(B) is the exponentiation of the coefficients.
Extraction to SPSS

		B	SE	Wald	df	Sig.	Exp(*B*)	Probability
Information transparency	Intersection	0.592	0.305	3.771	1	0.052		
	[Gender = 0]	0.055	0.427	0.017	1	0.897	1.057	0.514
	[Education_level = 0]	−0.256	0.531	0.232	1	0.630	0.774	0.436
	[Occupation = 0]	0.981	0.528	3.451	1	0.063	2.667	0.727
Training help	Intersection	−1.471	0.533	7.608	1	0.006		
	[Gender = 0]	−0.209	0.697	0.090	1	0.764	0.811	0.448
	[Education_level = 0]	0.252	0.782	0.104	1	0.748	1.286	0.563
	[Occupation = 0]	1.466	0.761	3.709	1	0.054	4.332	0.812

Table 9. Test for H_6: Positive_criteria
Non-statistically significant results are not included in the table.
B is the coefficient for the constant and Exp(B) is the exponentiation of the coefficients.
Extraction to SPSS

H_6: The null hypothesis is partially accepted. The occupation variable is statistically significant – with some degree of relaxation ($P < 0.10$) in the criterion for information transparency and help for employee training (Table 9). An unemployed investor is 2.667 and 4.332 times more likely to invest in SRI funds where transparency and help for training are positive investment criteria than an employed investor. This suggests that the probability of investing in SRI funds which promote information transparency and help for training increasing with unemployment. This may be because investors without work are more concerned about the destination of their investment. However, gender and education have no statistically significant explanatory power in explaining the social aspects that most concern investors in their investment process.

H_7: The variable 'gender' is statistically significant for criteria regarding activities that abuse the environment. The variable 'level of education' is statistically significant – with some degree of relaxation ($P < 0.10$) – for criteria regarding activities that abuse human and labour rights, lack of transparency and support for repressive and dictatorial regimes. Education plays a central role in making individuals incorporate societal values in their investment decisions (Table 10). Finally, the occupation variable is statistically significant for exclusion activities such as nuclear production (Table 10). These activities are valued more than 6.009 times more highly by unemployed respondents than by employed respondents.

Hypotheses H_8, H_9 and H_{12} relate to risk, performance and SRI trends in Spain. It can be seen that the gender, level of education and occupation variables are not statistically significant for H_8 and H_9.Thereby, the null hypothesis cannot be accepted.[2]

[2]As the results of the analysis are not statistically significant for these hypotheses, we have not included the SPSS results in the paper. For a more detailed analysis of the treatment of the data please contact the authors.

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		B	SE.	Wald	df	Sig.	Exp(*B*)	Probability
Activities that abuse and human and labour rights	Intersection	1.811	0.358	25.577	1	0.000		
	[Gender = 0]	−0.037	0.464	0.006	1	0.937	0.964	0.491
	[Education_level = 0]	−0.970	0.522	3.454	1	0.063	0.379	0.275
	[Occupation = 0]	0.679	0.580	1.367	1	0.242	1.971	0.663
Lack of transparency in business practices	Intersection	0.903	0.398	5.139	1	0.023		
	[Gender = 0]	−0.435	0.535	0.660	1	0.417	0.648	0.393
	[Education_level = 0]	−1.616	0.682	5.620	1	0.018	0.199	0.166
	[Occupation = 0]	0.985	0.649	2.305	1	0.129	2.678	0.728
Nuclear production	Intersection	−1.188	0.679	3.061	1	0.080		
	[Gender = 0]	−0.120	0.839	0.021	1	0.886	0.887	0.470
	[Education_level = 0]	−1.895	1.204	2.478	1	0.115	0.150	0.130
	[Occupation = 0]	1.793	0.914	3.853	1	0.050	6.009	0.857
Activities that abuse the environment	Intersection	−0.132	0.496	0.071	1	0.790		
	[Gender = 0]	−2.498	0.877	8.116	1	0.004	0.082	0.076
	[Education_level = 0]	−0.442	0.720	0.376	1	0.539	0.643	0.391
	[Occupation = 0]	2.074	0.736	7.930	1	0.005	7.955	0.888
Support for repressive or dictatorial regimes	Intersection	1.080	0.388	7.760	1	0.005		
	[Gender = 0]	−0.683	0.526	1.682	1	0.195	0.505	0.336
	[Education_level = 0]	−1.170	0.619	3.568	1	0.059	0.310	0.237
	[Occupation = 0]	0.946	0.636	2.214	1	0.137	2.576	0.720

Table 10. Test for H_7: Negative_criteria
Non-statistically significant results are not included in the table.
B is the coefficient for the constant and Exp(B) is the exponentiation of the coefficients.
Extraction to SPSS

		Estimate	SE	Wald	df	Sig
Threshold	[Tendency_SRI = 1]	−3.286	0.352	87.052	1	0.000
	[Tendency_SRI = 2]	−1.004	0.171	34.430	1	0.000
	[Tendency_SRI = 3]	1.720	0.192	80.330	1	0.000
Position	[Gender = 0]	0.510	0.216	5.597	1	0.018
	[Education_level = 0]	0.523	0.262	3.984	1	0.046
	[Occupation = 0]	0.120	0.237	0.255	1	0.613

Table 11. Test for H_{12}: Tendency_SRI
Extraction to SPSS

Meanwhile, the gender and educational level variables for H_{12} are statistically significant (Table 11). For this reason, the null hypothesis is partially accepted.

Conclusion

In this paper, we set out to study the perceptions held by the Spanish public regarding SRI criteria and real-life investment needs. We have conducted a field survey among 525 Spaniards.

The percentage of socially responsible individual investors in Spain is marginal (5.8% of the total sample). Most of these investors are women, in late middle age, highly educated, with middle and higher incomes. A certain lack of awareness was observed regarding SR financial products. The sustainability ratings agencies and sustainability

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indices are the SRI market instruments least known by the Spanish respondents. On the other hand, ethical banks are the most widely known SRI market instruments.

According to the results of the logistic regression test, the occupational and educational variables are the most important determinants when making investment decisions. In this sense, a higher level of education generates a more informed society, which enhances the development of SRI.

As a result of our research we propose some aspects that could be improved in order to correct the main weakness to the growth of SR financial products in Spain, which are: (i) failure to publicise the existence of these products; (ii) a perception of low returns; (iii) a lack of adaptation of these products to the needs of investors; and (iv) the limited development of SRI in Spain.

To tackle these problems, on the one hand, we propose some recommendations for institutions and managers:

1. Integrating sustainability into the business strategies of banks and financial institutions. SRI managers must formulate a specific SRI policy according to the investor's needs. A specific policy will facilitate the generation of socially responsible products in accordance with the demand and specifications of clients.
2. SRI managers must create products and generate an economic return that reflects the social and environmental values of investors. This implies carrying out market research to detect the demands of socially responsible investors.
3. Investors must be informed in detail about the characteristics of these products. SRI managers must make public information about methodologies and sources used for the ethical screening, the variety of products available, their financial, ethical and risk performance, etc.
4. Commercial brochures could be designed according to different investor preferences. Commercial brochures should highlight a specific set of criteria in order to appeal to each specific cluster.
5. It is necessary a greater promotion of SRI by the government. Some of the possible actions could introduce some SRI regulations to improve transparency and stimulate demand or some fiscal incentives, following the example from other European countries.
6. It is necessary for future works to examine specific SRI demands from different types of investors and to compare them with supply in order to explain why the retail SRI market is not growing.

On the other hand, we propose some recommendations for citizens and investors too.

1 Investors should demand more information from their managers about the different types of financial products that they can offer.
2 Better integration of shareholder engagement practices in investment funds is necessary. Currently, there is considerable interest in shareholder engagement in sustainability issues because, among other reasons, shareholders can stimulate significant improvements in corporate performance on ESG issues, thereby enhancing and protecting shareholder value, particularly over the longer term (Gifford, 2006).
3 People should be more informed about SRI. It is therefore necessary that schools and universities teach about alternative strategies to the traditional investment process.

Nevertheless, despite these drawbacks, there is an existing basis for growth (58% of the sample believes that SR investments will grow in the future).

Future work will seek to extend the study of the perception of SRI from Spain to a European level in order to evaluate the degree of integration enjoyed by these products and instruments in various European societies.

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